Chemistry Detailed Practice Workbook

Concepts • Numericals • Reasoning • Mini Mock

Part 1 — Foundation Practice

• Calculate the moles present in 18 g of water.

• Find the number of neutrons in an atom with A = 23 and Z = 11.

• Write the electron configuration of a simple main-group atom with atomic number 17.

• State the shape expected for a central atom with four bonding pairs and no lone pair.

• Convert 0.5 mol in 250 mL solution into molarity.

Part 2 — Gas & Thermodynamics

• A gas occupies 2 L at 1 atm. What volume is expected at 2 atm if temperature is unchanged?

• A 0.20 mol ideal-gas sample is at a known temperature and pressure. Set up the equation needed to find its volume.

• Classify a process as exothermic or endothermic from the sign of ΔH.

• Use Hess's law conceptually to combine two reactions into a target reaction.

• Explain why absolute temperature is used in gas-law equations.

Part 3 — Equilibrium, Redox & Electrochemistry

• Identify oxidation and reduction in a simple electron-transfer equation.

• Predict qualitatively how increasing reactant concentration affects an equilibrium system.

• Identify the anode and cathode in a galvanic cell.

• Explain why oxidation and reduction always occur together in a redox reaction.

• Design a two-step method for balancing a redox equation: oxidation-state method or half-reaction method.

Part 4 — Organic Chemistry

• Identify the functional group in alcohols, aldehydes, ketones, carboxylic acids and amines.

• Differentiate saturated and unsaturated hydrocarbons.

• Predict whether a simple alkene reaction is addition or substitution.

• Explain the difference between sigma and pi bonding.

• Give one reason benzene undergoes substitution rather than simple addition under many common conditions.

Part 5 — Mini Mock

• 1. Which law connects V and T at constant pressure? A) Boyle B) Charles C) Hess D) Faraday

• 2. What is the molarity of 1 mol solute in 1 L solution? A) 0.1 M B) 0.5 M C) 1 M D) 10 M

• 3. Oxidation occurs at: A) anode B) cathode C) salt bridge D) electrolyte only

• 4. A pi bond is formed mainly by: A) head-on overlap B) sidewise overlap C) ionic attraction D) proton transfer

• 5. A catalyst primarily: A) increases ΔH B) changes equilibrium constant C) lowers activation energy D) changes atomic number

• Answers: 1-B, 2-C, 3-A, 4-B, 5-C.